

Supplementary Module 6 | Deep spatial architecture and mechanism-deepening evidence

A

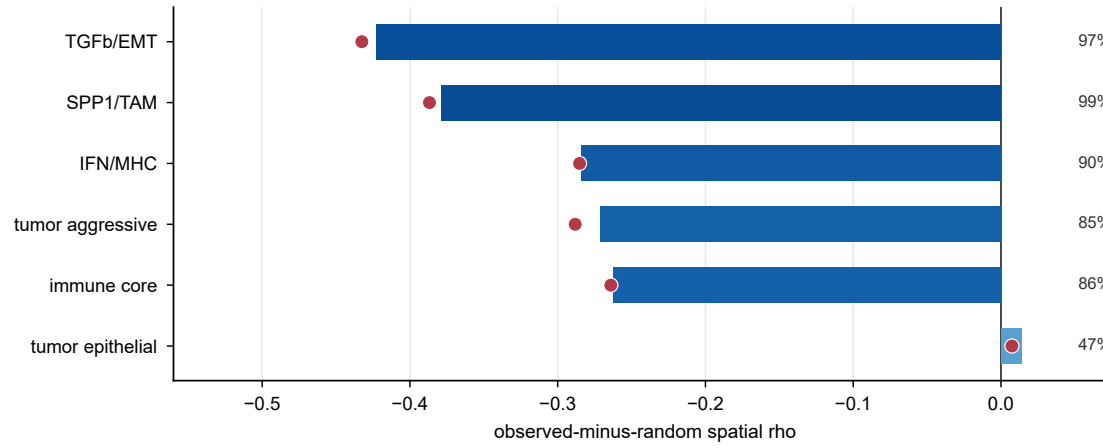
Deep spatial architecture evidence chain



Claim level: spatial association and prioritization; perturbation is still required for causality.

B

CAF-myeloid core specificity



red dot: observed rho; bar: observed - random

H

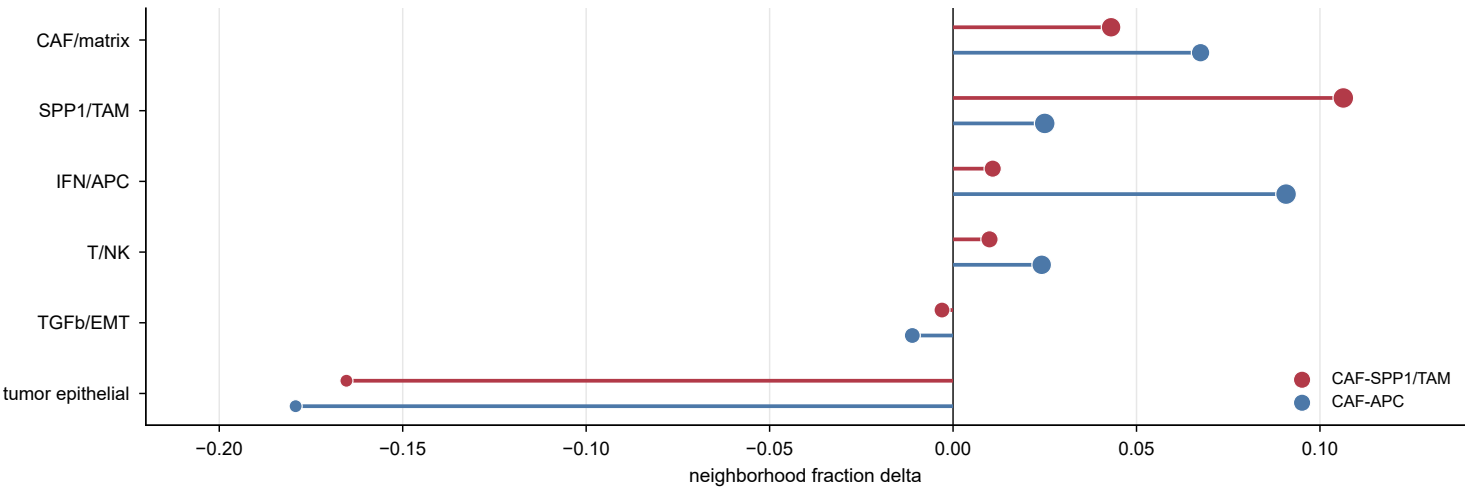
Claim boundary

- S** specific spatial architecture
- S** cell-level CAF-myeloid proximity
- S** interface gradient organization
- M** geometry tunes coupling
- O** causal signaling direction

Use as deepening evidence, not standalone causal proof.

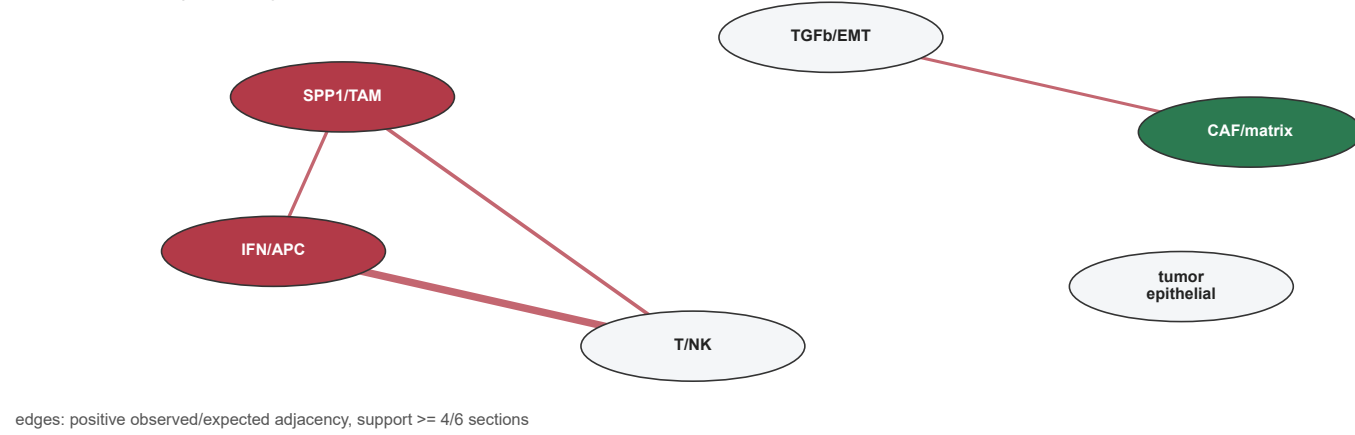
C

Xenium CAF-domain neighborhoods



D

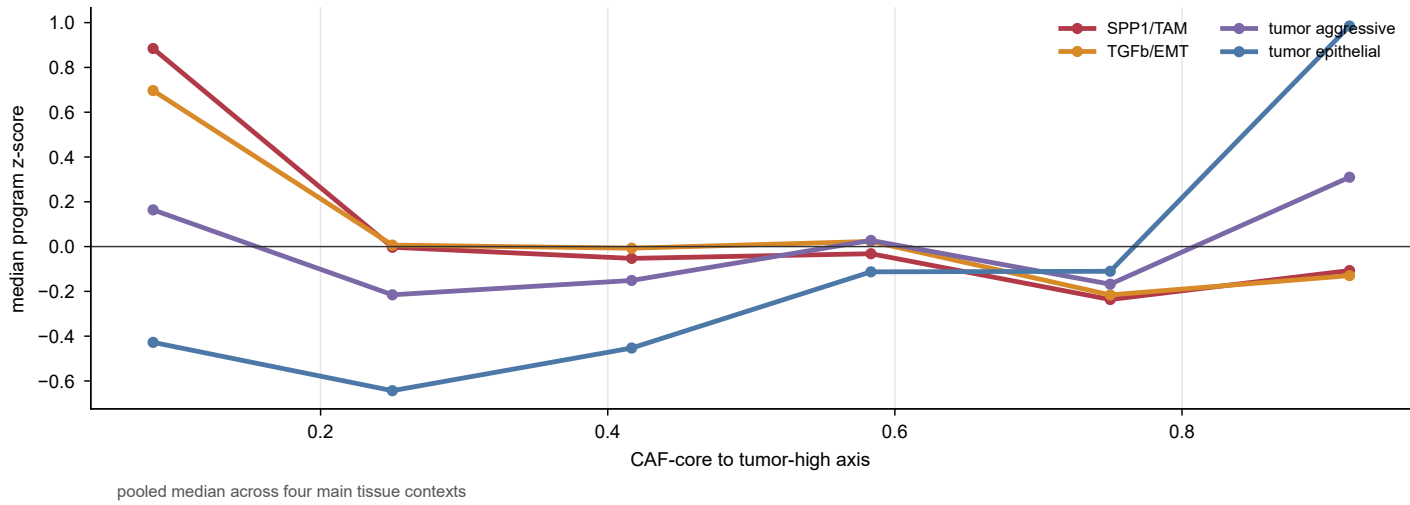
Cell-state adjacency network



edges: positive observed/expected adjacency, support >= 4/6 sections

E

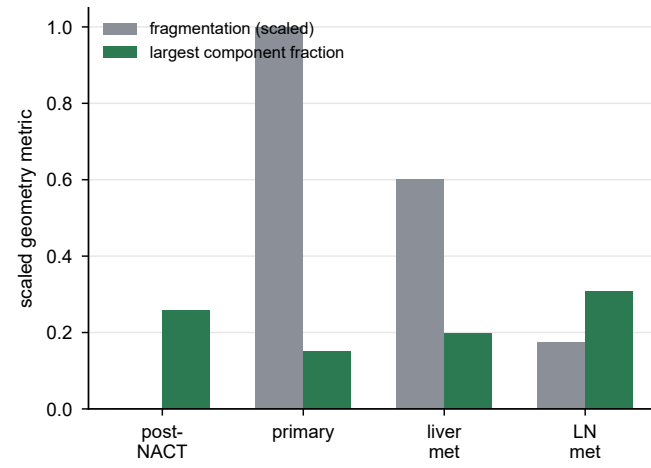
Core-to-interface transition



pooled median across four main tissue contexts

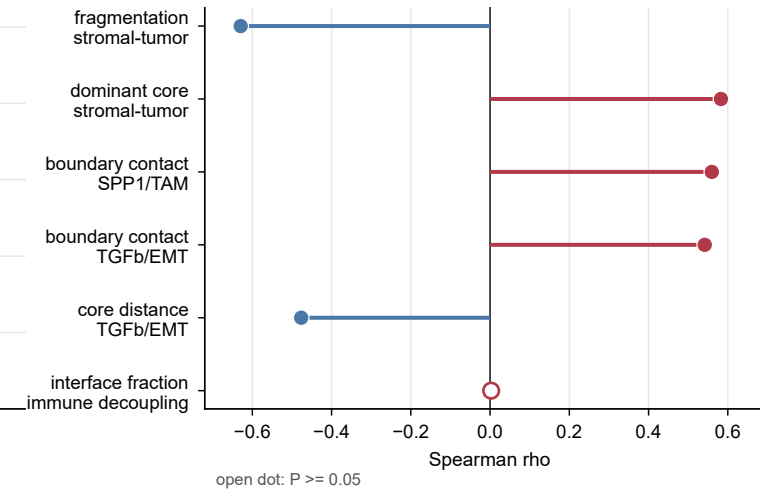
F

CAF-core geometry by context



G

Geometry-biological associations



open dot: P >= 0.05

Interpretation: ED30-ED33 converge on a deeper model in which CAF-myeloid niches are spatially specific, reproduced at cell-neighborhood resolution, organized along a core-to-interface axis and modulated by CAF-core geometry.

Manuscript use: describe this as spatial architecture and perturbation-priority evidence. Reserve causal language for future blocking, organoid/co-culture or perturb-seq style validation.